

# Internet and Cybersecurity Basics

---

## Lesson Objectives

By the end of this lesson, students should be able to:

1. Describe how the Internet works using the **client-server model**.
  2. Explain the purpose of **HTTP** and **DNS** in online communication.
  3. Understand the principles and benefits of **cloud computing**.
  4. Identify common **cybersecurity threats** and methods to prevent them.
  5. Create and document a **Digital Safety Checklist** for students.
- 

## 1. How the Internet Works

The **Internet** is a global network that connects millions of devices using communication protocols. It allows the exchange of data such as text, audio, and video between computers.

### The Client-Server Model

This is the foundation of how online communication happens.

- **Client:** A device or program that requests services (e.g., web browser, mobile app).
- **Server:** A powerful computer that stores and delivers data to clients.

### Example:

When you visit [www.wikipedia.org](http://www.wikipedia.org):

1. Your browser (**client**) sends a request to the **Wikipedia server**.
2. The server finds the page and sends the data back.
3. Your browser displays the content.

This communication happens in milliseconds and is powered by **network protocols** such as TCP/IP (Transmission Control Protocol / Internet Protocol).

---

## 2. HTTP and DNS

### HTTP (HyperText Transfer Protocol)

HTTP defines how data is formatted and transmitted between web clients and servers.

- **HTTP Request:** The browser asks for information.
- **HTTP Response:** The server sends the requested data.
- **HTTPS:** A secure version that encrypts communication using SSL/TLS, ensuring privacy and data integrity.

#### Example of an HTTP request:

GET /index.html HTTP/1.1  
Host: www.example.com

### DNS (Domain Name System)

Humans remember names, but computers use numbers (IP addresses). DNS acts like the Internet's **phonebook**, converting website names into IP addresses.

#### Example:

www.google.com → 142.250.190.78

Without DNS, you would have to remember numeric IP addresses instead of easy-to-read names.

---

## 3. Cloud Computing Overview

### Definition

Cloud computing is the delivery of computing services — servers, storage, databases, networking, software — over the Internet (“the cloud”).

## Service Models

| Model | Description                                               | Example                   |
|-------|-----------------------------------------------------------|---------------------------|
| IaaS  | Infrastructure as a Service – renting servers and storage | Amazon Web Services (AWS) |
| PaaS  | Platform as a Service – hosting for developers            | Google App Engine         |
| SaaS  | Software as a Service – use software online               | Gmail, Microsoft 365      |

## Benefits of Cloud Computing

- **Scalability:** Easily increase or reduce resources.
- **Accessibility:** Access files anywhere, anytime.
- **Cost efficiency:** Pay only for what you use.
- **Reliability:** Automatic data backups and recovery.

**Example:** Using Google Drive to store documents and access them from any device.



## 4. Introduction to Cybersecurity

### Definition

Cybersecurity involves protecting systems, networks, and programs from digital attacks or unauthorized access.

### Common Threats

| Threat     | Description                                 | Example              |
|------------|---------------------------------------------|----------------------|
| Virus      | Self-replicating program that corrupts data | File infector virus  |
| Phishing   | Fraudulent emails that trick users          | Fake bank login page |
| Ransomware | Locks data and demands payment              | WannaCry attack      |

|                     |                                                   |                     |
|---------------------|---------------------------------------------------|---------------------|
| <b>Spyware</b>      | Tracks user activity secretly                     | Keyloggers          |
| <b>Trojan Horse</b> | Disguises as useful software but contains malware | Fake antivirus apps |

## Cybersecurity Protection Measures

### Firewalls

- Software or hardware that filters traffic between your device and the Internet.
- Prevents unauthorized access.

### Encryption

- Converts readable data into coded form.
- Used in online transactions, emails, and file storage.

### Backups

- Keep a second copy of important files on another drive or cloud.
- Protects against data loss due to hardware failure or attacks.

### Authentication and Passwords

- Use **strong passwords** (mix of uppercase, lowercase, digits, and symbols).
- Implement **two-factor authentication (2FA)** for added security.

### Software Updates

- Regularly update your OS and applications to patch security vulnerabilities.

## Mini Project: Digital Safety Checklist

### Task

Create a **Digital Safety Checklist** that educates students on how to stay safe online.

## Your Checklist Should Include:

1. Keep your operating system and apps updated.
2. Use strong and unique passwords.
3. Enable two-factor authentication.
4. Avoid sharing personal information online.
5. Don't click on unknown links or attachments.
6. Backup your data regularly.
7. Use antivirus or anti-malware software.
8. Lock your devices when unattended.
9. Avoid using public Wi-Fi for banking.
10. Be cautious of phishing and scam emails.

## Deliverable

- Write your checklist in a word processor or design it in Canva.
- Save as **Digital\_Safety\_Checklist.pdf**
- Upload it to your GitHub repository under /Week9\_Cybersecurity/.



## Git Task

1. Create a new branch named week9\_cybersecurity.
2. Add your **Digital Safety Checklist** and a short summary (README.md).
3. Commit changes with a message:
4. git commit -m "Added Week 9 Cybersecurity Checklist"
5. Push your branch to GitHub.



## Exercise

### Question:

If you visit [www.schoolportal.com](http://www.schoolportal.com) and your browser cannot find it, what might be the cause?

- a) Faulty DNS
- b) No Internet connection
- c) Server is down
- d) All of the above

**Answer:** ☒ d) All of the above

---

